

Question:

To analyze a complete login workflow using Burp Suite.

- 1. Open Burp Suite and start intercepting traffic.**
- 2. Visit a website that you own, a training lab, or a practice application.**
- 3. Capture the following requests:**
 - **Home Page Request**
 - **Login Page Request**
 - **Login Submission Request**
 - **Logout Request**
- 4. For each request identify:**
 - **Request Method**
 - **URL**
 - **Parameters**
 - **Headers**
 - **Cookies**
 - **Response Code**
- 5. Send all captured requests to Repeater.**
- 6. Analyze:**
 - **What changes after login?**
 - **Which cookie is responsible for maintaining the session?**
 - **Which parameters are sent to the server?**
 - **What information does the response contain?**
- 7. Create a flow diagram showing:**

Browser → Request → Server → Response → Login Success

8. Document your findings with screenshots.

Submission:

Prepare a PDF containing:

- **Screenshots of all requests**
- **Screenshots of responses**
- **Session cookie details**
- **Login flow explanation**
- **Flow diagram**

Objective

To analyze a complete login workflow using Burp Suite by intercepting HTTP requests and responses, identifying session cookies, and understanding the authentication process.

Background

Burp Suite is a web application security testing tool used for intercepting, analyzing, and modifying HTTP/HTTPS traffic between a browser and a web server. It helps security professionals and students understand how web applications communicate and manage user authentication.

In this experiment, Burp Suite Community Edition was used to capture and analyze the login workflow of a practice web application provided by PortSwigger Web Security Academy.

Tools Used

- Burp Suite Community Edition
- Mozilla Firefox Browser
- PortSwigger Web Security Academy Lab

Website Used

PortSwigger Web Security Academy Practice Lab

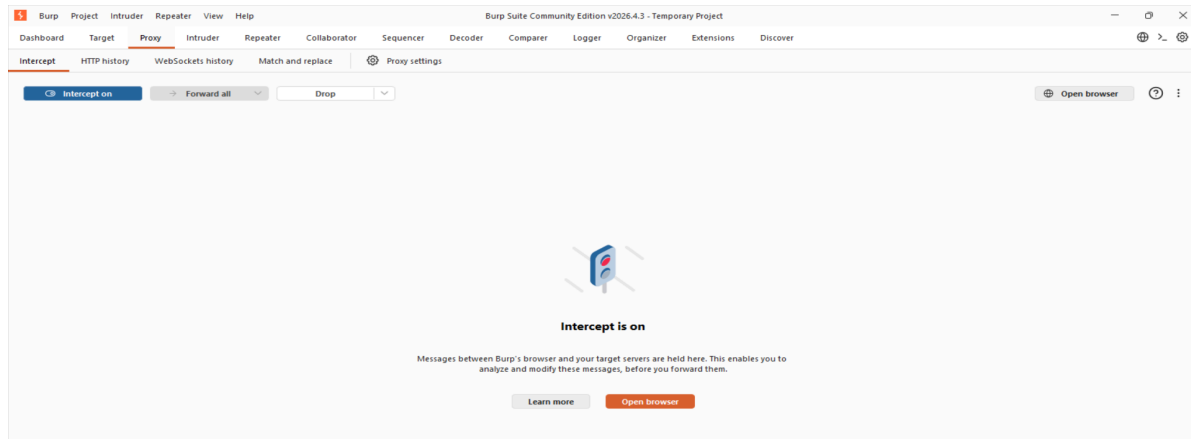
Procedure

Step 1: Starting Burp Suite

Burp Suite Community Edition was launched and the proxy intercept option was enabled from: Proxy → Intercept

Intercept was turned ON to capture browser requests.

Screenshot:



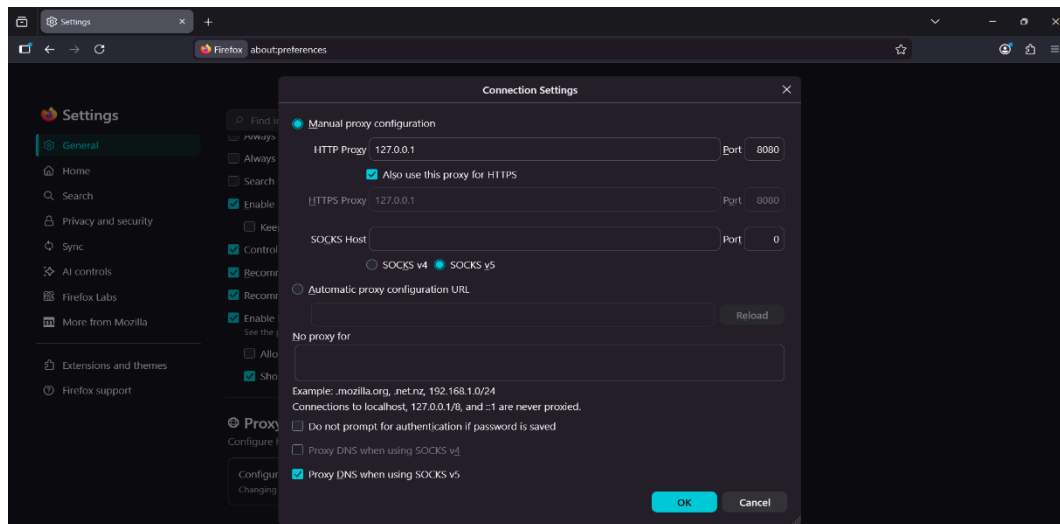
Step 2: Configuring Browser Proxy

Mozilla Firefox was configured to use Burp Suite proxy settings:

IP Address: 127.0.0.1

Port: 8080

Screenshot:



Step 3: Opening the PortSwigger Lab

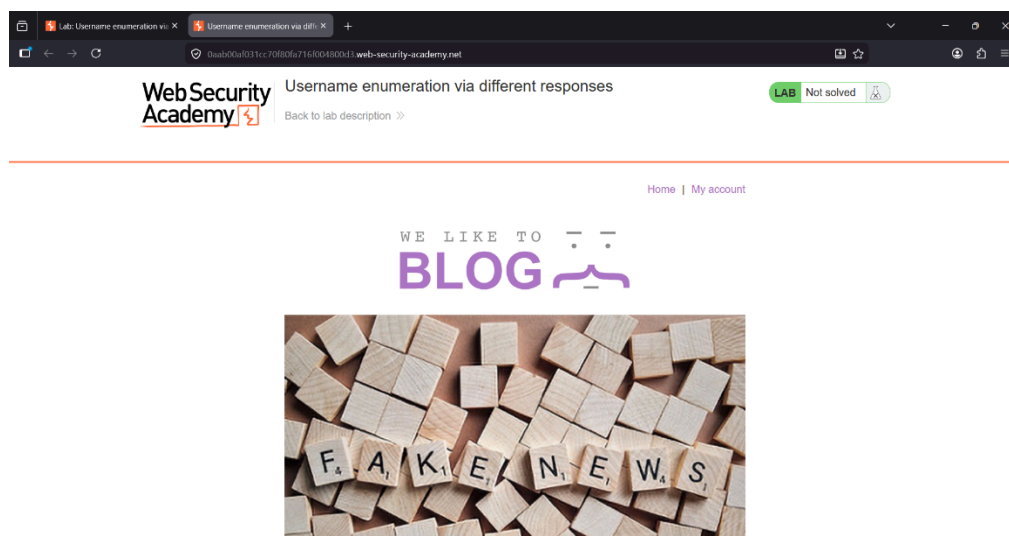
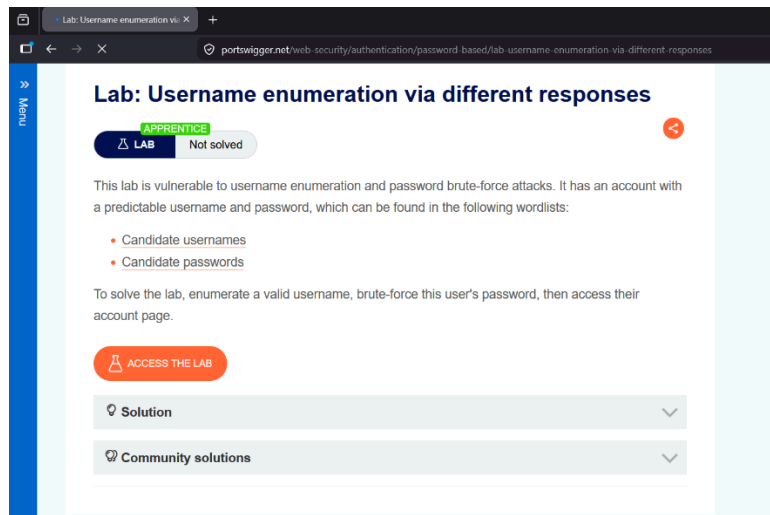
- Opened the PortSwigger Web Security Academy website.
- Navigated to Authentication Labs.
- Selected a practice lab.

- Clicked:

Access the Lab

- The practice website opened in the browser.

Screenshot:



Step 4: Capturing Home Page Request

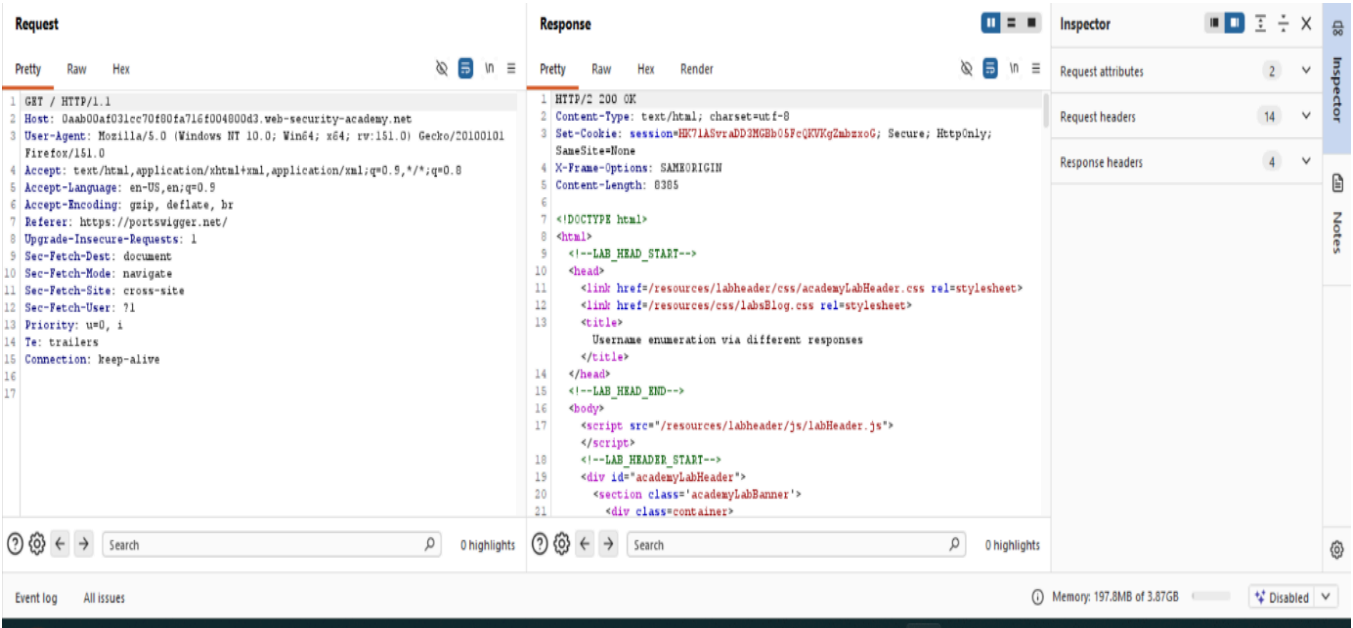
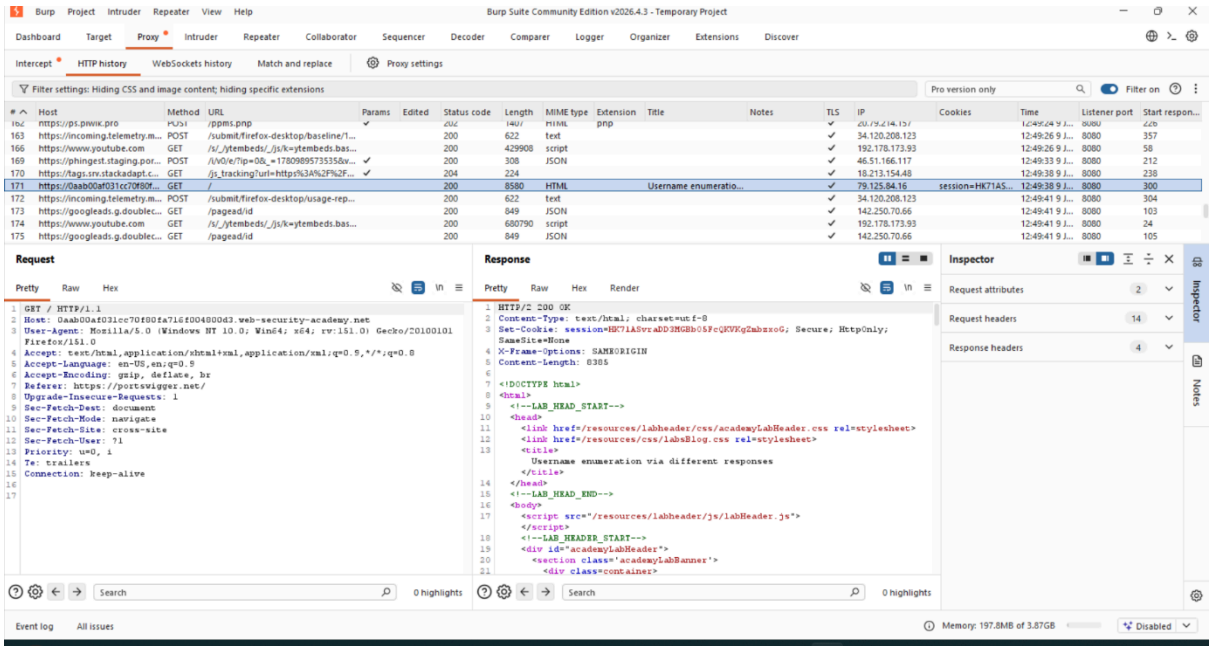
The practice lab homepage was opened in Firefox. Burp Suite intercepted the HTTP request successfully.

Home Page Request: GET / HTTP/2

Observations

Parameter	Value
Method	GET
Response Code	200
Purpose	Requesting homepage

Screenshot:



Step 5: Capturing Login Page Request

The “My Account” option was clicked to open the login page.

Login Page Request: POST /login HTTP/2

Observations

Parameter	Value
Method	POST
Response Code	200
Purpose	Opening login form

The following credentials were entered in the login form:

Username: wiener

Password: peter

After clicking the login button, Burp Suite intercepted the login submission request.

Captured Login Request

POST /login HTTP/2

Host: 0aab00af031cc70f80fa716f004800d3.web-security-academy.net

Cookie: session=HK71ASvraDD3MGBbO5FcQKVKgZmbzxoG

Content-Type: application/x-www-form-urlencoded

username=wiener&password=peter

Screenshot:

The screenshot displays the Burp Suite Community Edition v2026.4.3 interface. The top menu bar includes options like Dashboard, Target, Proxy, Intruder, Repeater, Collaborator, Sequencer, Decoder, Comparer, Logger, Organizer, Extensions, and Discover. The main window is divided into several panes. The 'HTTP history' pane shows a list of intercepted requests, with the selected request (ID 234) being a POST to /login. The 'Request' pane shows the raw HTTP request, including headers like Host, Cookie, User-Agent, and Accept, and the body with username=wiener and password=peter. The 'Response' pane shows the raw HTTP response, including headers like Content-Type, X-Frame-Options, and Content-Length, and the body which is an HTML page. The 'Inspector' pane on the right shows the 'Request headers' section, listing various headers and their values.

#	Host	Method	URL	Params	Status code	Length	MIME type	Extension	Title	Notes	TLS	IP	Cookies	Time	Listener port	Start respon...
226	https://www.gitatic.com	GET	/youtube/mg/lotie/subscribe_acti...		200	4320	JSON	json			✓	142.250.67.227		12:53:36 9...	8080	43
227	https://moodle-https.fastly-	POST	/		200	765					✓	199.232.105.91		12:53:36 9...	8080	307
228	https://0aab00af031cc70f90f...	GET	/login		200	3287	HTML		Username enumeratio...		✓	79.125.84.16		12:53:36 9...	8080	179
230	https://incoming.telemetry...	POST	/submit/firefox-desktop/newtab/1/...		200	622	text				✓	34.120.208.123		12:53:41 9...	8080	316
231	https://0aab00af031cc70f90f...	GET	/academyLabHeader		101	147					✓	79.125.84.16		12:53:41 9...	8080	204
232	https://googleads.g.doublec...	GET	/pagead/id		200	849	JSON				✓	142.250.146.154		12:53:54 9...	8080	86
233	https://googleads.g.doublec...	GET	/pagead/id		200	849	JSON				✓	142.250.146.154		12:53:54 9...	8080	83
234	https://0aab00af031cc70f90f...	POST	/login		200	3352	HTML		Username enumeratio...		✓	34.246.129.62		12:55:57 9...	8080	203

Request

```
1 POST /login HTTP/2
2 Host: 0aab00af031cc70f90f716f004800d3.web-security-academy.net
3 Cookie: session=HK71ASvraD0INGb05FcQVFPg2abx06
4 User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:151.0) Gecko/20100101 Firefox/151.0
5 Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8
6 Accept-Language: en-US,en;q=0.8
7 Accept-Encoding: gzip, deflate, br
8 Content-Type: application/x-www-form-urlencoded
9 Content-Length: 30
10 Origin: https://0aab00af031cc70f90f716f004800d3.web-security-academy.net
11 Referer: https://0aab00af031cc70f90f716f004800d3.web-security-academy.net/login
12 Upgrade-Insecure-Requests: 1
13 Sec-Fetch-Dest: document
14 Sec-Fetch-Mode: navigate
15 Sec-Fetch-Site: same-origin
16 Sec-Fetch-User: ?1
17 Priority: u=0, i
18 Te: trailers
19
20 username=wiener&password=peter
```

Response

```
1 HTTP/2 200 OK
2 Content-Type: text/html; charset=utf-8
3 X-Frame-Options: SAMEORIGIN
4 Content-Length: 3244
5
6 <!DOCTYPE html>
7 <html>
8 <!--LAB_HEAD_START-->
9 <head>
10 <link href="/resources/labheader/css/academyLabHeader.css" rel="stylesheet">
11 <link href="/resources/css/labs.css" rel="stylesheet">
12 <title>
13 Username enumeration via different responses
14 </title>
15 </head>
16 <!--LAB_HEAD_END-->
17 <body>
18 <script src="/resources/labheader/js/labHeader.js">
19 </script>
20 <!--LAB_HEADER_START-->
21 <div id="academyLabHeader">
22 <section class="academyLabHeader">
23 <div class="container">
24 <div class="logo">
25 </div>
26 <div class="title-container">
27 <h2>
```

Inspector

Name	Value
:scheme	https
:method	POST
:path	/login
:authority	0aab00af031cc70f90f...
cookie	session=HK71ASvraD...
user-agent	Mozilla/5.0 (Windows...
accept	text/html,application/...
accept-language	en-US,en;q=0.9
accept-encoding	gzip, deflate, br
content-type	application/x-www-f...
content-length	30
origin	https://0aab00af031c...
referer	https://0aab00af031c...
upgrade-insecure-re...	1
sec-fetch-dest	document
sec-fetch-mode	navigate
sec-fetch-site	same-origin

Response headers

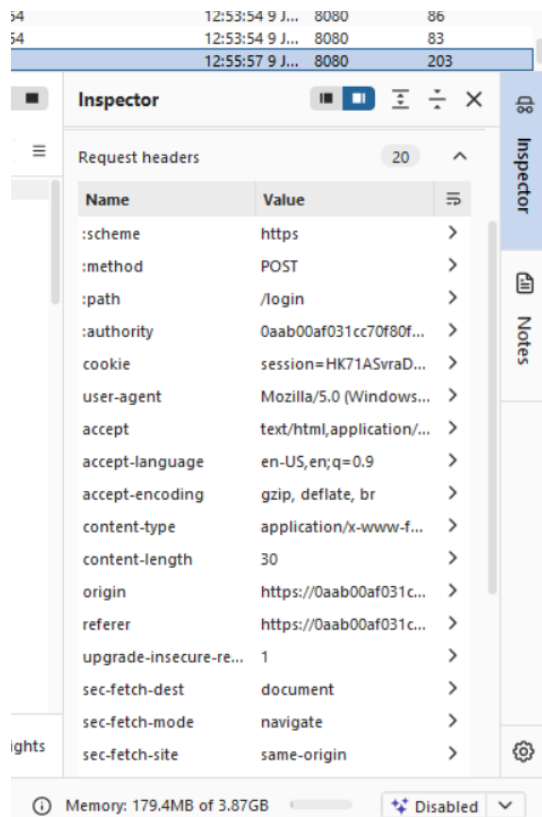
Name	Value
Content-Type	text/html; charset=ut...
X-Frame-Options	SAMEORIGIN
Content-Length	3244

Request body parameters

Name	Value
username	wiener
password	peter

Request attributes

Name	Value
Protocol	HTTP/1
Method	POST
Path	/login



Analysis of Login Submission Request

Request Method

POST:-POST method was used to securely transmit credentials to the server.

URL:- /login

This endpoint handled user authentication.

Parameters Sent to Server

Parameter Description

username Username entered by user

password Password entered by user

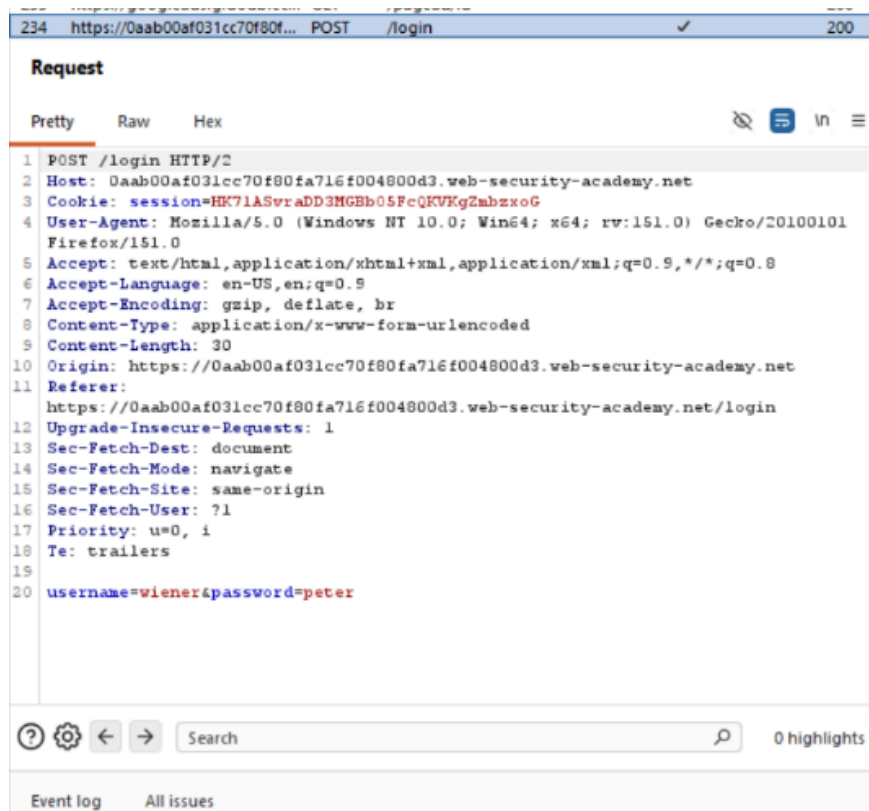
Parameters identified:

- username=wiener
- password=peter

Important Headers

Header	Purpose
Host	Identifies target server
Cookie	Maintains session
Content-Type	Specifies form data format
Referer	Indicates previous webpage
User-Agent	Browser information

Screenshot:

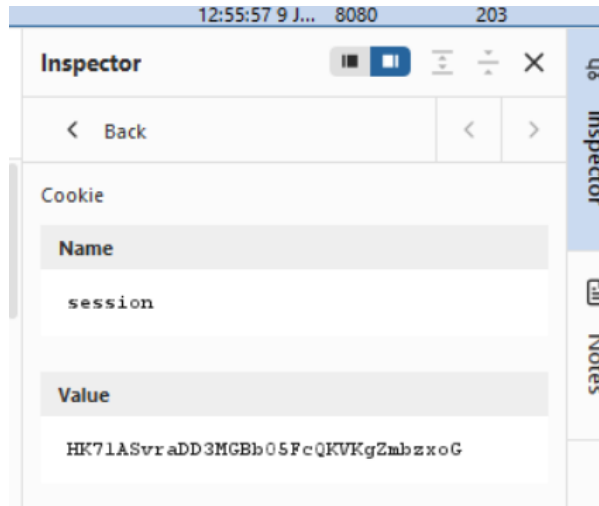


Session Cookie Identified

session=HK71ASvraDD3MGBbO5FcQKVKgZmbzxoG

This session cookie maintained communication between browser and server.

Screenshot:



Step 6: Login Response Analysis

After forwarding the login request, the server generated a response.

Captured Response

HTTP/2 200 OK

Content-Type: text/html

Response Analysis

Parameter	Observation
Response Code	200 OK
Response Type	HTML
Authentication Result	Login Failed
Session Cookie	Present

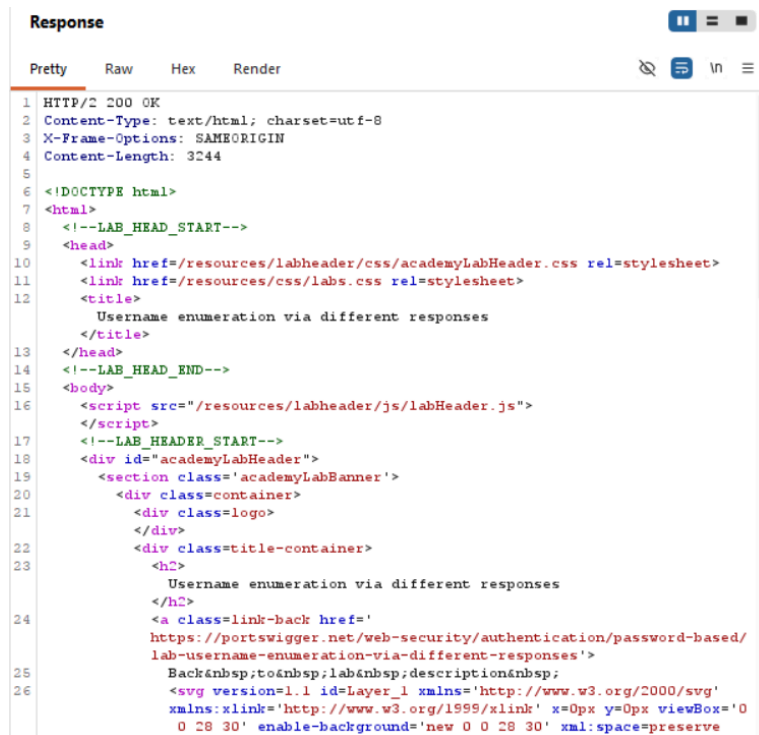
Response Body Observation

The response contained an authentication failure message:

Invalid username or password

This indicated that the server validated user credentials before granting access.

Screenshot:



The screenshot shows a web browser response in a tool like Burp Suite. The response is in 'Pretty' view and shows an HTTP 200 OK status. The content type is text/html. The response body is an HTML document with a title 'Username enumeration via different responses'. The body contains a script tag for 'resources/labheader/js/labHeader.js' and a div with id 'academyLabHeader'. Inside this div, there is a section class 'academyLabBanner' containing a container with a logo and a title container with an h2 heading 'Username enumeration via different responses'. Below the heading, there is a link-back href='https://portswigger.net/web-security/authentication/password-based/lab-username-enumeration-via-different-responses'. The link-back href is followed by a description 'Back to lab' and a description 'Username enumeration via different responses'. The response also includes an SVG element with version 1.1, id 'Layer_1', and a xlink href 'http://www.w3.org/2000/svg'. The SVG element has a width of 0px, a height of 0px, and a viewBox of '0 0 28 30'. The SVG element is styled with 'enable-background: new 0 0 28 30' and 'xml:space:preserve'.

```
1 HTTP/2 200 OK
2 Content-Type: text/html; charset=utf-8
3 X-Frame-Options: SAMEORIGIN
4 Content-Length: 3244
5
6 <!DOCTYPE html>
7 <html>
8 <!--LAB_HEAD_START-->
9 <head>
10 <link href=/resources/labheader/css/academyLabHeader.css rel=stylesheet>
11 <link href=/resources/css/labs.css rel=stylesheet>
12 <title>
13 Username enumeration via different responses
14 </title>
15 </head>
16 <!--LAB_HEAD_END-->
17 <body>
18 <script src=/resources/labheader/js/labHeader.js>
19 </script>
20 <!--LAB_HEADER_START-->
21 <div id=academyLabHeader>
22 <section class=academyLabBanner>
23 <div class=container>
24 <div class=logo>
25 </div>
26 <div class=title-container>
27 <h2>
28 Username enumeration via different responses
29 </h2>
30 <a class=link-back href=
31 https://portswigger.net/web-security/authentication/password-based/
32 lab-username-enumeration-via-different-responses'>
33 Back to lab
34 </a>
35 <div class=description>
36 Username enumeration via different responses
37 </div>
38 </div>
39 </div>
40 </div>
41 </div>
42 </div>
43 </div>
44 </div>
45 </div>
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90 </div>
91 </div>
92 </div>
93 </div>
94 </div>
95 </div>
96 </div>
97 </div>
98 </div>
99 </div>
100 </div>
```

Step 7: Sending Request to Repeater

The captured login request was analyzed further using Burp Repeater.

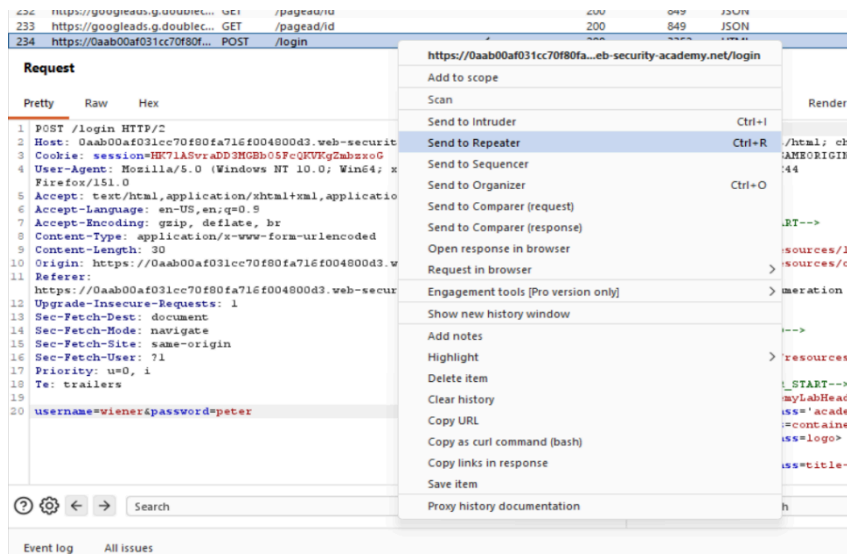
Procedure

- Right-clicked login request
- Selected:

Send to Repeater

- Opened Repeater tab
- Modified request parameters
- Sent modified requests to server

Screenshot:



Repeater Analysis

Different usernames and passwords were tested.

Example:

- username=admin
- password=peter

Observations

- Different credentials generated different responses.
- Response lengths changed.
- Authentication behavior changed based on parameters.
- Session cookie remained active.

Screenshot:

Response

PrettyRawHexRender

1HTTP/2 200 OK

2Content-Type: text/html; charset=utf-8

3X-Frame-Options: SAMEORIGIN

4Content-Length: 3244

5

6<!DOCTYPE html>

7<html>

8<!--LAB_HEAD_START-->

9<head>

10<link href=/resources/labheader/css/academyLabHeader.css rel=

11stylesheet>

12<link href=/resources/css/labs.css rel=stylesheet>

13<title>

14Username enumeration via different responses

15</title>

16</head>

17<!--LAB_HEAD_END-->

18<body>

19<script src=/resources/labheader/js/labHeader.js>

20</script>

21<!--LAB_HEADER_START-->

22<div id=academyLabHeader>

23<section class=academyLabBanner>

24<div class=container>

25<div class=logo>

26</div>

27<div class=title-container>

28<h2>

29Username enumeration via different responses

30</h2>

31<a class=link-back href=

32https://portswigger.net/web-security/authentication/password-based/lab-username-enumeration-via-di

33fferent-responses>

34

35</div>

36</section>

37</div>

38</body>

39</html>

0 highlights

Inspector

NameValue

:schemehttps

:methodPOST

:path/login

:authority0ae3007903c7386581...

cookie

session=58Uolia6Hlfl...

user-agentMozilla/5.0 (Windows...

accepttext/html,application/...

accept-languageen-US,en;q=0.9

accept-encodinggzip, deflate, br

content-typeapplication/x-www-f...

content-length29

originhttps://0ae3007903c7...

refererhttps://0ae3007903c7...

upgrade-insecure-re...1

sec-fetch-destdocument

sec-fetch-modenavigate

sec-fetch-sitesame-origin

sec-fetch-user?1

priorityu=0, i

te trailers

Inspector

Request attributes2

ProtocolHTTP/1HTTP/2

NameValue

MethodPOST

Path/login

Request query parameters0

Request body parameters2

NameValue

usernameadmin

passwordpeter

Response headers	
Name	Value
Content-Type	text/html; charset=ut...
X-Frame-Options	SAMEORIGIN
Content-Length	3244

Step 8: Logout Request Analysis

After login workflow analysis, logout functionality was analyzed. The logout button on the application was clicked, and Burp Suite intercepted the request.

Screenshot:

The screenshot displays the Burp Suite interface with the following components:

- Intercepted Requests Table:**

#	Host	Method	URL	Params	Edited	Status code	Length	MIME type	Extension	Title	Notes	TLS	IP	Cookies	Time	Listener port	Start respon...
708	https://phingest.staging.por...	POST	/v0/e/tp=0&_w=1780993243866&v...		✓	200	308	JSON				✓	13.226.244.8	>sessionid=CfD8F...	13:50:40.9 J...	8080	13:51
709	https://login.portswigger.net	GET	/v2/logout?client_id=F1PHGMosque...		✓	302	672	HTML				✓	104.18.34.171		13:50:44.9 J...	8080	262
710	https://stripe.com	POST	/b		✓	200	320	text				✓	54.187.119.242		13:50:45.9 J...	8080	309
- Request Details (GET /logout):**

```

HTTP/2 302 Found
Date: Tue, 09 Jun 2026 08:20:52 GMT
Content-Type: text/html; charset=utf-8
Content-Length: 52
Location: https://portswigger.net/
CF-Ray: a08ab344c0b7e08-BOM
CF-Cache-Status: DYNAMIC
Cache-Control: no-store, max-age=0, no-transform
Server: cloudflare
Strict-Transport-Security: max-age=31536000; includeSubDomains
Vary: Accept
Pragma: no-cache
X-Auth0-Id: 0.04C
X-Auth0-Request-Id: 4b4e4c236c07b9365b9f
X-Content-Type-Options: nosniff
X-Ratelimit-Limit: 300
X-Ratelimit-Remaining: 296
X-Ratelimit-Reset: 1780993254
Alt-Svc: h3="443"; ma=66400

```
- Inspector Panel:**
 - Request attributes:** Protocol: HTTP/2, HTTP/2
 - Request query parameters:** (Empty)
 - Request cookies:**

Name	Value
Sessionid	CfD8F8F8MeYQV04Vps...
did	sv0:7ccb4ff-9556-46...
auth0	sy9WY3i6315dppgll...
did_compat	sv0:7ccb4ff-9556-46...
auth0_compat	sy9WY3i6315dppgll...
_gcl_ae	1.1.1940490571.17809...
_cf_bm	SYOKTIEQ_uXamefaUL...
ph_phc_zmmwVwifm...	{'session_id': '019eab...
 - Request headers:** (24 headers listed)

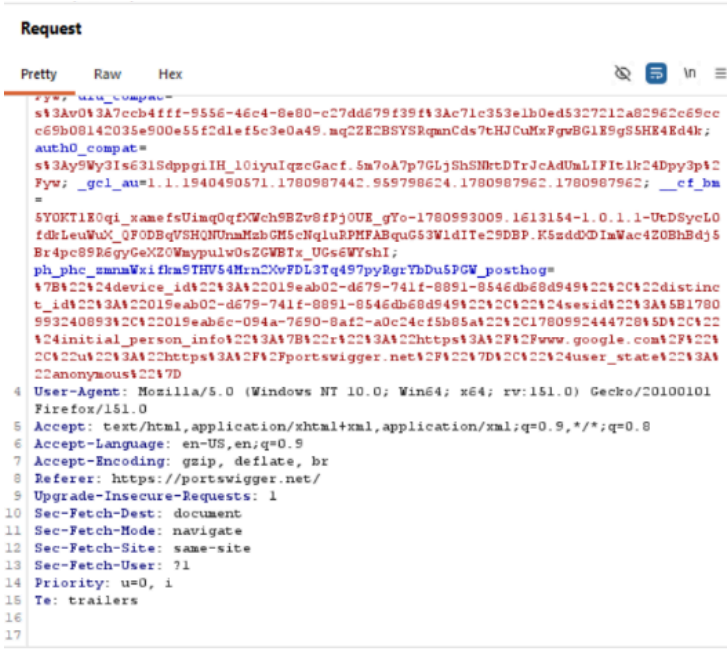
Captured Logout Request

GET /logout HTTP/2

Host: 0aab00af031cc70f80fa716f004800d3.web-security-academy.net

Cookie: session=HK71ASvraDD3MGBbO5FcQKVKGZmbzxoG

Screenshot:



Logout Response

HTTP/2 302 Found

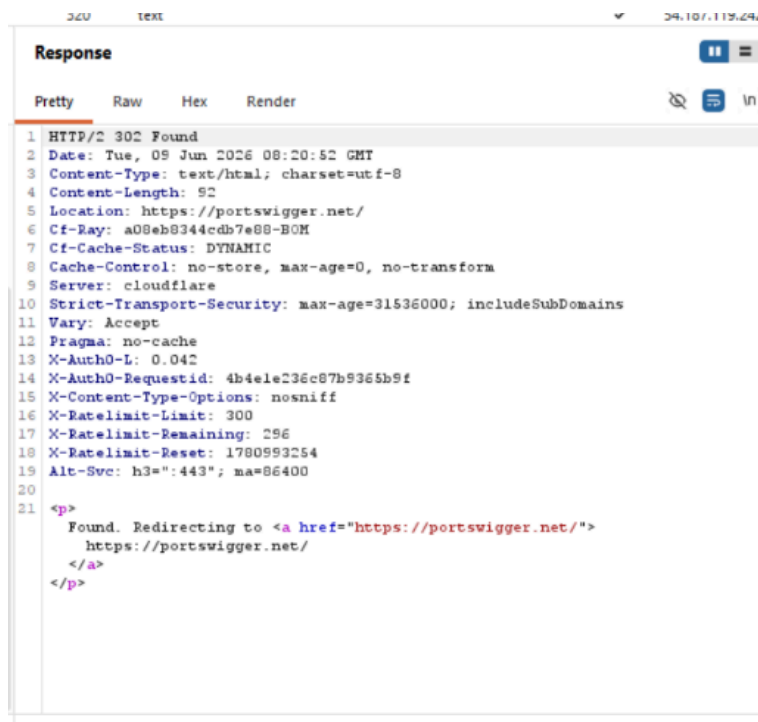
Location: /

Set-Cookie: session=; expires=Thu, 01 Jan 1970 00:00:00 GMT

Logout Response Analysis

Parameter	Observation
Response Code	302 Found
Redirect Location	Homepage
Session Status	Expired
Authentication Status	Logged Out

Screenshot:



Session Management Analysis

The application used session cookies to maintain authentication state between browser and server.

Session Cookie

session=HK71ASvraDD3MGBbO5FcQKVKgZmbzxoG

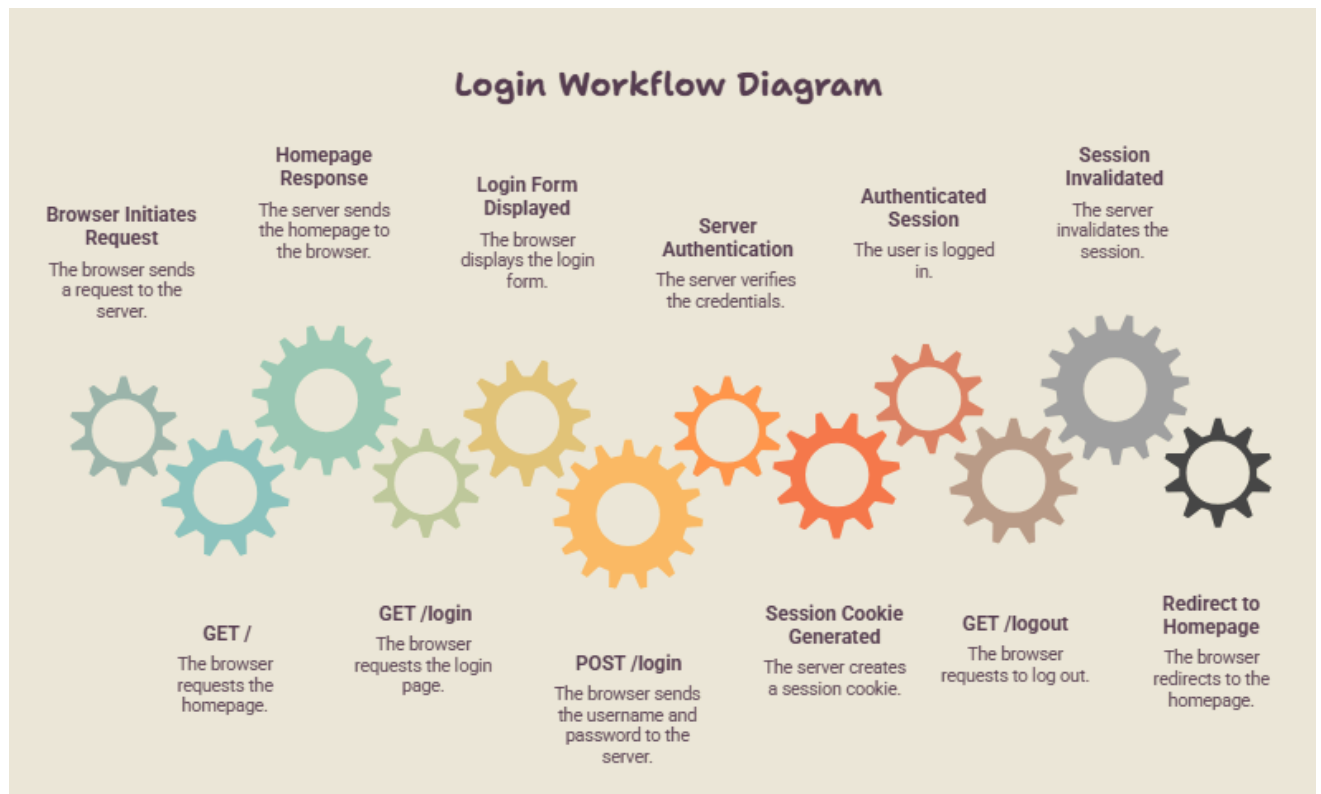
Purpose of Session Cookie

- Maintains user session
- Tracks authentication state
- Identifies client requests
- Enables persistent communication

Screenshot:

Request cookies			8	^
Name	Value			
SessionId	CfDJ8FIMeYQV04Vlps...	>		
did	s:v0:7ccb4fff-9556-46...	>		
auth0	s:y9Wy3ls631Sdppgil...	>		
did_compat	s:v0:7ccb4fff-9556-46...	>		
auth0_compat	s:y9Wy3ls631Sdppgil...	>		
_gcl_au	1.1.1940490571.17809...	>		
__cf_bm	5Y0KT1EOqi_xamefsUi...	>		
ph_phc_zmnmWxifkm...	{"\$device_id": "019eab...	>		
Request headers			24	^

Login Workflow Diagram



Findings

- Burp Suite successfully intercepted HTTP requests and responses.
- GET and POST methods were analyzed.
- Login credentials were transmitted using POST requests.
- Session cookies maintained user sessions.
- Repeater allowed modification and testing of requests.
- Logout functionality properly invalidated session cookies.

Conclusion

This experiment successfully demonstrated how Burp Suite can be used to analyze complete login workflows in web applications. Different HTTP requests, responses, headers, parameters, cookies, and session management techniques were analyzed using the PortSwigger practice lab. The experiment provided practical understanding of authentication workflows and web application communication mechanisms.